VELTRAXX'26

इलेक्ट्रॉनिकी एवं सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF ELECTRONICS AND INFORMATION TECHNOLOGY
Chips to Startup Programme
INSTITUTION'S INNOVATION COUNCIL (Ministry of Education Initiative)
BAJAJ INSTITUTE OF TECHNOLOGY, WARDHA

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TEAM NAME: QuadraX

PROBLEM STATEMENT NO: PS 16

PROBLEM STATEMENT TITLE: Open Domain-Specific Silicon / Edge Hardware Innovation (Wildcard)

PS Category: Hardware

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Problem Statement:

REAL-WORLD PROBLEM

- Edge IoT devices need AES-128 encryption and decryption in hardware.
- Software AES on Cortex-M4: ~1,136 cycles per 128-bit block.
- Physical attacks can target cryptographic keys through fault injection.

WHY IT MATTERS

- Slow software crypto limits secure communication on battery devices.
- A fault-leaked key can compromise an entire device fleet.
- Closed crypto silicon / EDA flows increase development cost.

CURRENT CHALLENGES

- Separate encrypt/decrypt datapaths increase area.
- Fault detection and rapid key wiping must be tightly coupled.
- The accelerator must integrate cleanly with a standard SoC bus.

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QuadraX AES-128 Cryptographic SoC for Edge Silicon:

ONE-LINE SOLUTION

• Resource-shared bidirectional AES-128 SoC with 1-cycle tamper zeroization, AXI4-Lite integration and FPGA verification.

HOW IT SOLVES THE PROBLEM

• One datapath supports encryption + decryption using a shared composite GF(2^8) S-Box/InvS-Box.
• Tamper detection → 1-cycle abort + atomic wipe of 1,408 round-key bits, master key and outputs.
• Unmaskable security_irq reports an active security fault.
• AXI4-Lite register file: CTRL, STATUS, KEY0–3, DIN0–3, DOUT0–3.
• 10 cycles/block at 50 MHz; 9/9 regression tests passed.

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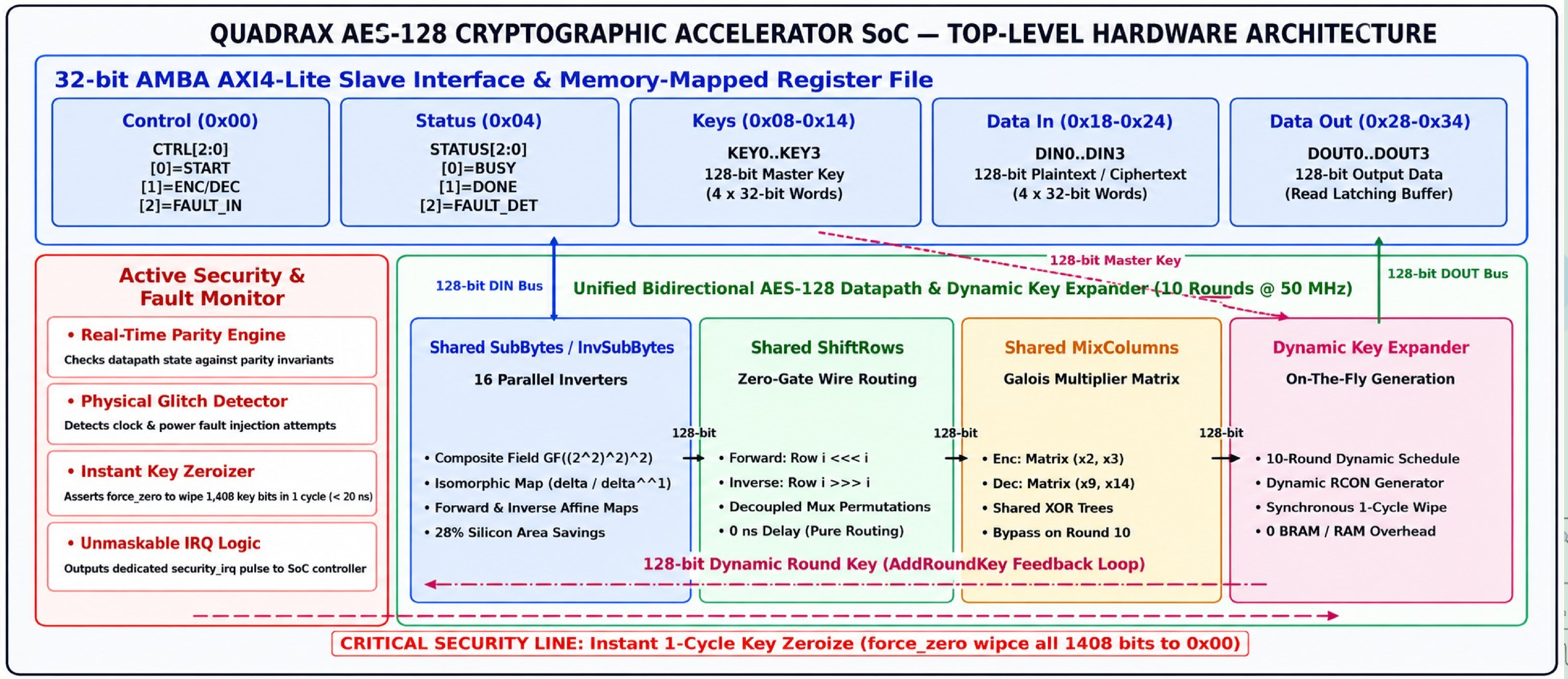
ARCHITECTURAL BLOCK DIAGRAM:



PURPOSE OF BLOCKS :

INPUT — AXI4-LITE SLAVE + REGISTER FILE

• Receives key, data and control commands from a bus master.
• Standard interface makes the accelerator reusable SoC IP.

PROCESSING — BIDIRECTIONAL AES-128 CORE

• **One shared round datapath for forward and inverse transforms.**
• Shared S-Box, ShiftRows and MixColumns; on-the-fly key expansion.

OUTPUT — DOUT REGISTERS + STATUS / LEDS

• **Ciphertext / recovered plaintext is read back through AXI.**
• Hardware demo: 0x97 encrypt • 0x2A decrypt • 0x00 tamper wipeout.

ARCHITECTURAL TRADE-OFFS

• **Iterative vs unrolled: lower area; cost = 10 cycles/block.**
• Shared S-Box vs two tables: smaller substitution logic; added mux/affine logic.
• Parity tamper check vs redundant core: low overhead; narrower fault coverage.
• AXI4-Lite vs full AXI: simpler integration; no burst transfers.

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OUTPUTS AND RESULTS:

LIVE FPGA DEMONSTRATION — BASYS3 / ARTIX-7 @ 50 MHz

- Encryption: 0x97 — PASS.
- Decryption: 0x2A — PASS.
- Tamper / zeroization: 0x00 — PASS.

PERFORMANCE

- Latency: 10 cycles = 200 ns per 128-bit block.
- Throughput: 0.64 Gbps.

IMPLEMENTATION

- Yosys 0.9 gate-level netlist committed.
- RTL, simulation and FPGA evidence are included in the project repository.
- SkyWater 130 nm ASIC physical flow is future work; current delivered scope is RTL + synthesis + simulation + FPGA.

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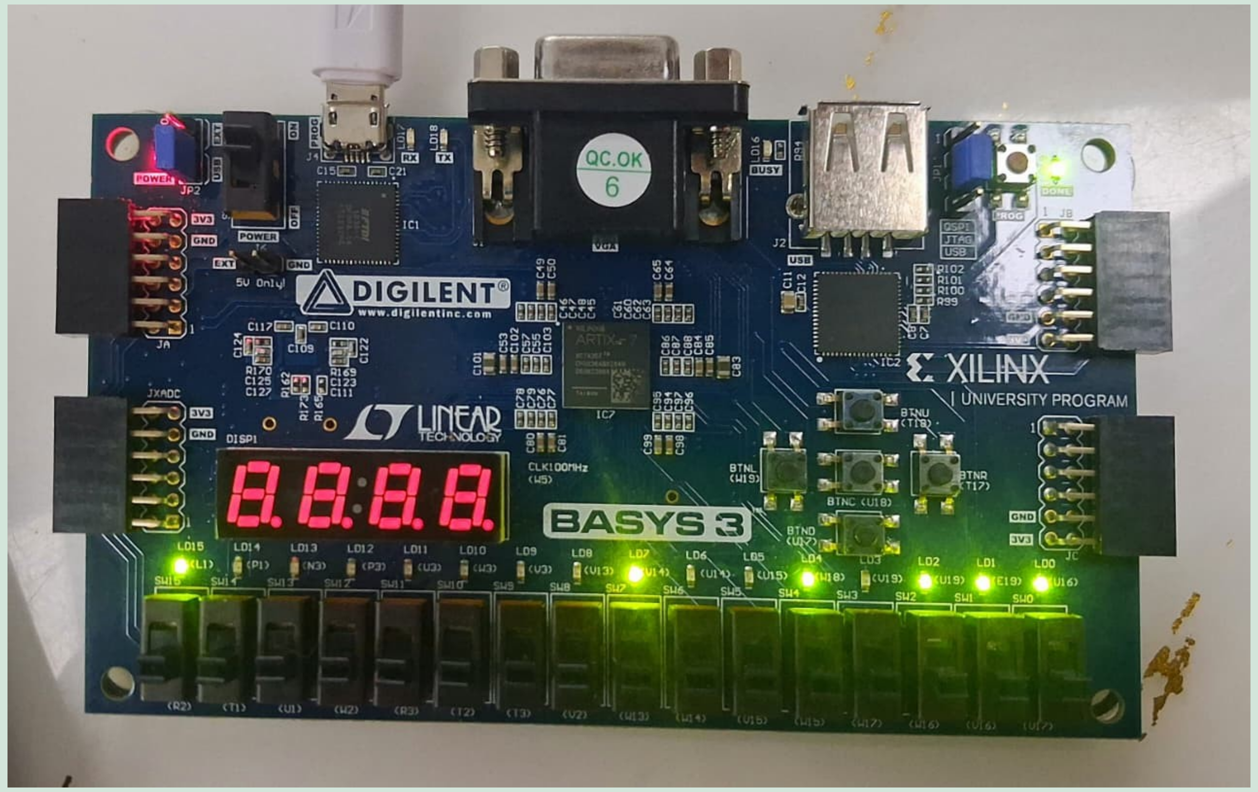
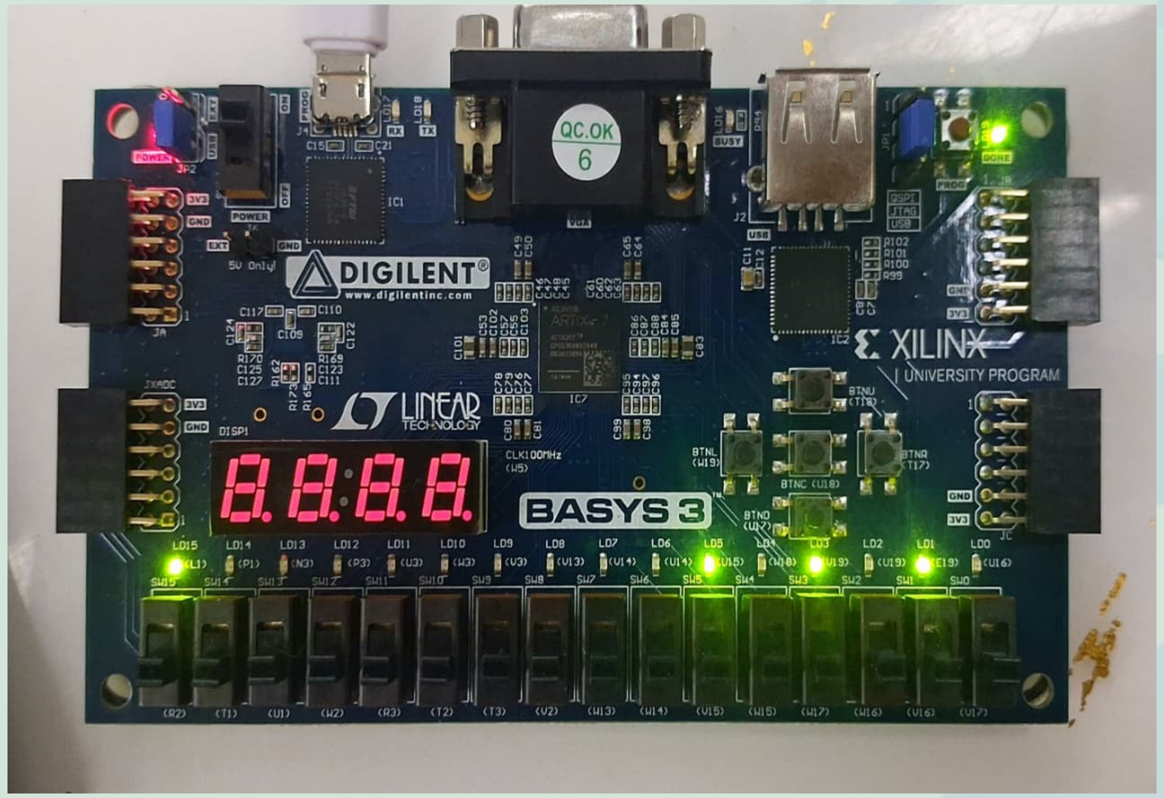
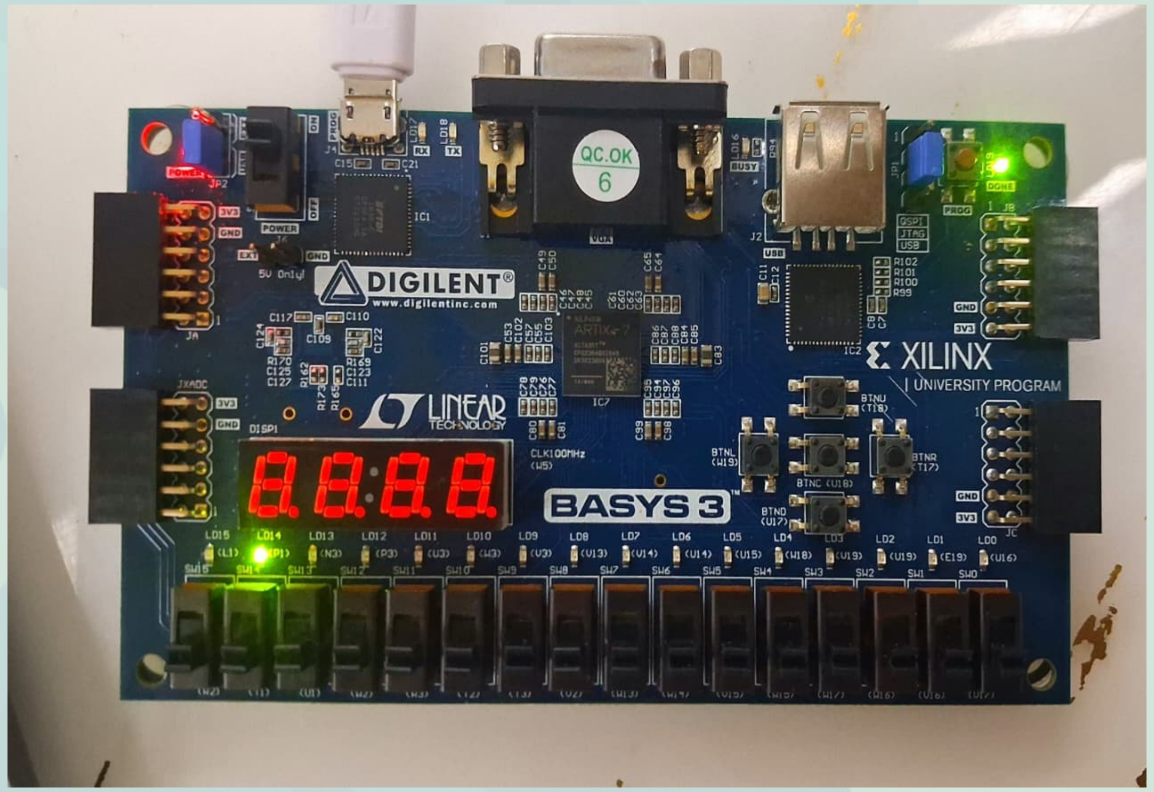


NIST Enc (0x97 PASS)



NIST Dec (0x2A PASS)



Tamper Wipe (0x00 PASS)

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OUTPUTS AND RESULTS :

VERIFICATION

- 9/9 regression tests PASS: 4 encrypt KAT + 4 decrypt KAT + fault/zeroization.
- 3/3 architectural directives PASS: bidirectional core, tamper/zeroize + IRQ, AXI4-Lite.
- Fault injection: STATUS = 0x0000000C; security_irq asserted → PASS.

NIST KNOWN-ANSWER TEST

- TV1 encryption ciphertext: 3AD77BB40D7A3660A89ECAF32466EF97 → PASS.
- TV1 decryption plaintext recovered → PASS.

EVIDENCE

- Self-checking Verilog testbenches + GTKWave waveform evidence committed in the repository.

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RESEARCH AND REFERENCES :

• NIST — FIPS 197, Advanced Encryption Standard (AES).
• NIST SP 800-38A — block-cipher test vectors.
• ARM — AMBA AXI4-Lite Interface Specification.
• mbed TLS Cortex-M4 benchmark used for the software baseline.
• Yosys Open Synthesis Suite; Icarus Verilog; AMD Vivado.

PROJECT SOURCES :

• VELTRAXX26-PS16-QuadraX — architecture, RTL, verification and FPGA evidence.
• AES-Crypto-SoC — AES-128 SoC and RTL-to-GDSII implementation work.

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